B2.1 Membranes and membrane transport (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 STATE the effect of cholesterol on membrane fluidity at low temperature. [B2.1.12] [1]

2 DISTINGUISH between phagocytosis and pinocytosis. [B2.1.13] [2]

3 STATE the number of sodium ions and potassium ions moved in one cycle of the sodium-potassium pump, and the direction of each. [B2.1.15] [2]

4 NAME the group of cell-adhesion molecules that usually join cells to the extracellular matrix. [B2.1.17] [1]

5 OUTLINE the role of tight junctions between intestinal epithelium cells. [B2.1.17] [2]

6 NAME the neurotransmitter that opens the channel of the nicotinic receptor. [B2.1.14] [1]

Section B: Understanding

15 MARKS

7 EXPLAIN how the fatty acid composition of phospholipids affects membrane fluidity. [B2.1.11] [3]

8 EXPLAIN how the structure of cholesterol determines its position in the bilayer. [B2.1.12] [2]

9 **EXPLAIN** why exocytosis depends on the fluidity of membranes. [B2.1.13] [2]

10 **OUTLINE** how the sodium-potassium pump uses ATP to move ions. [B2.1.15] [3]

11 **EXPLAIN** why glucose uptake by sodium-dependent cotransporters is called indirect active transport. [B2.1.16] [2]

12 **COMPARE** voltage-gated and neurotransmitter-gated ion channels. [B2.1.14] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

13 [B2.1.11, B2.1.12]

Groups of carp were kept at 5, 15 or 25 °C for four weeks. The phospholipids in their liver cell membranes were then analysed.

WATER TEMPERATURE / °C	UNSATURATED FATTY ACIDS / % OF TOTAL	CHOLESTEROL / % OF MEMBRANE LIPID
5	68	17
15	59	20
25	49	24

a **DESCRIBE** the relationship between water temperature and the percentage of unsaturated fatty acids. [1]

b **CALCULATE** the percentage increase in unsaturated fatty acids from 25 °C to 5 °C. [2]

c **EXPLAIN** the advantage to the carp of the change in fatty acids at 5 °C. [2]

d **SUGGEST** why the cholesterol content is higher at 25 °C. [2]

e Membranes from the 5 °C and 25 °C groups were both tested at 15 °C. **PREDICT** which would be more fluid. [1]

f **SUGGEST** why the change in membrane composition takes days, not minutes. [2]

Section D: Longer answer

6 MARKS

14 **EXPLAIN** how cells lining the small intestine absorb glucose against its concentration gradient. [6]

[B2.1.15, B2.1.16]